PROGRAMMING WITH PYTHON

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Computer Applications

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Python Basics and Standard Library

Comprehensions - List

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List Comprehensions

- **List comprehensions** in Python can be used to generate more varied sequences.
- It is a method to create lists using rules.
- It offers a shorter syntax when you want to create a new list based on the values of an existing list.

- l1 = ['hello' for x in range(5)]
  - This statement would create a list containing hello five times.
- The previous statement is equivalent to the following piece of code.
  - l1 = []
  - for x in range(5):
  - l1.append(‘hello’)

- The general form of list comprehension is

  - [<expr> for <variable> in <iterable>]

- Semantically, this is equivalent the following.
    - Create an empty list.
    - Execute the for part of the list comprehension.
    - Evaluate the expr each time.
    - Append that to the list.
    - The result is the list so created.

List Comprehensions

```
>>> l2 = [ x * x for x in range(5)]

>>> l2
[0, 1, 4, 9, 16]
>>> type(l2)
<class 'list'>
```

- Conditional List comprehension
  - new_list = [expression for variable in iterable if condition == True]

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
squares_of_evens = [x**2 for x in numbers if x % 2 == 0]
print(squares_of_evens)
```

```
[4, 16, 36, 64, 100]
```

- Nested list Comprehension
  - [expression for element in outer_list for sub_element in inner_list]
- Nested list comprehension can also have conditions
  - [expression for item1 in iterable1
  - if condition1
  - for item2 in iterable2 if condition2]

Nested List Comprehensions

- ```
  matrix = [
      [1, 2, 3],
      [4, 5, 6],
      [7, 8, 9]
  ]
  flattened = [num for row in matrix for num in row]
  print(flattened)
  ```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
matrix = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
]
even_numbers = [num for row in matrix for num in row if num % 2 == 0]
print(even_numbers)
```

```
[2, 4, 6, 8]
```
```

Recap

- List Comprehensions
  - Simple
  - Conditional
  - Nested
  - Nested Conditional

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THANK YOU

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